

2. Voice Control Muto Motion

Before running this program, you need to bind the port numbers of the voice board and the ROS expansion board on the host machine. You can refer to the previous chapter for binding. When entering the Docker container, you need to mount this voice board so that it can be recognized within the container.

1. Program Function Description

After the program starts, say "Hello, yahboom" to the module. The module will reply with "I'm here" to wake up the voice board. Then you can issue voice commands to control Muto's basic movements, such as moving forward, backward, turning left/right. The movement will stop after 5 seconds.

2. Program Code Reference Path

After entering the Docker container, the source code for this function is located at:

```
/root/yahboomcar_ros2_ws/yahboomcar_ws/src/yahboomcar_voice_ctrl/yahboomcar_voice_ctrl/Voice_Ctrl_driver.py
```

3. Program Startup

3.1. Startup Command

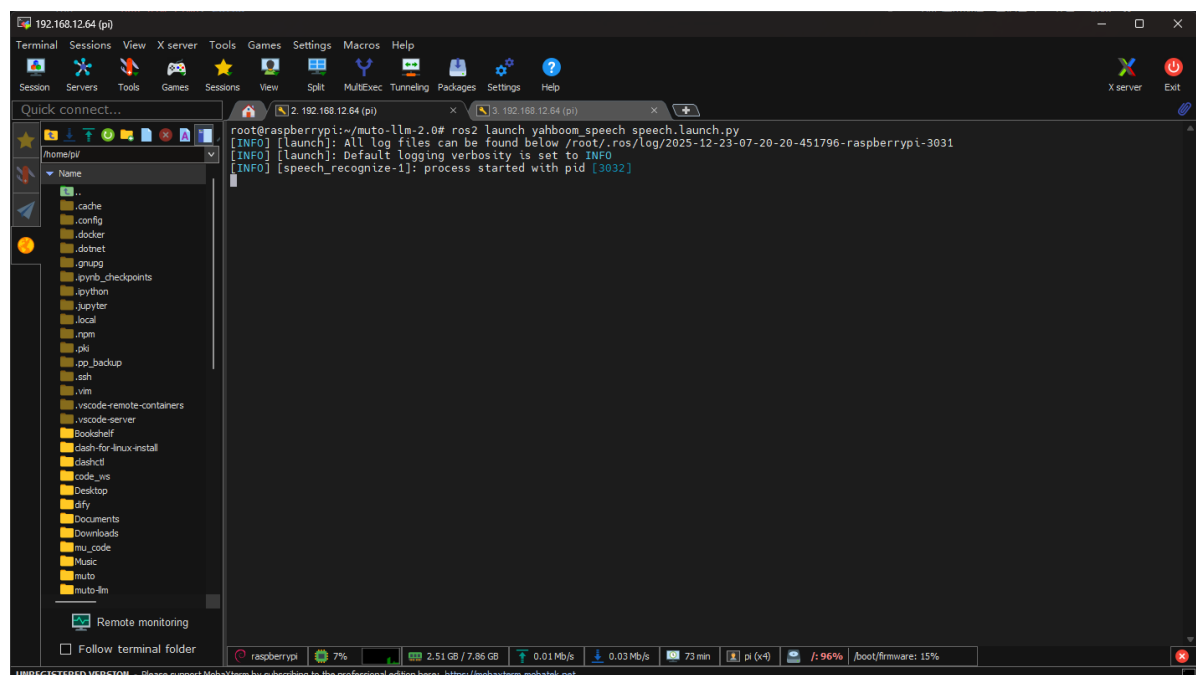
Open two terminals and enter the Docker container in each.

If there is no running Docker container, use the `start_muto_docker` script to start one.

If a container is already running, use the `enter_muto_docker` script to enter it.

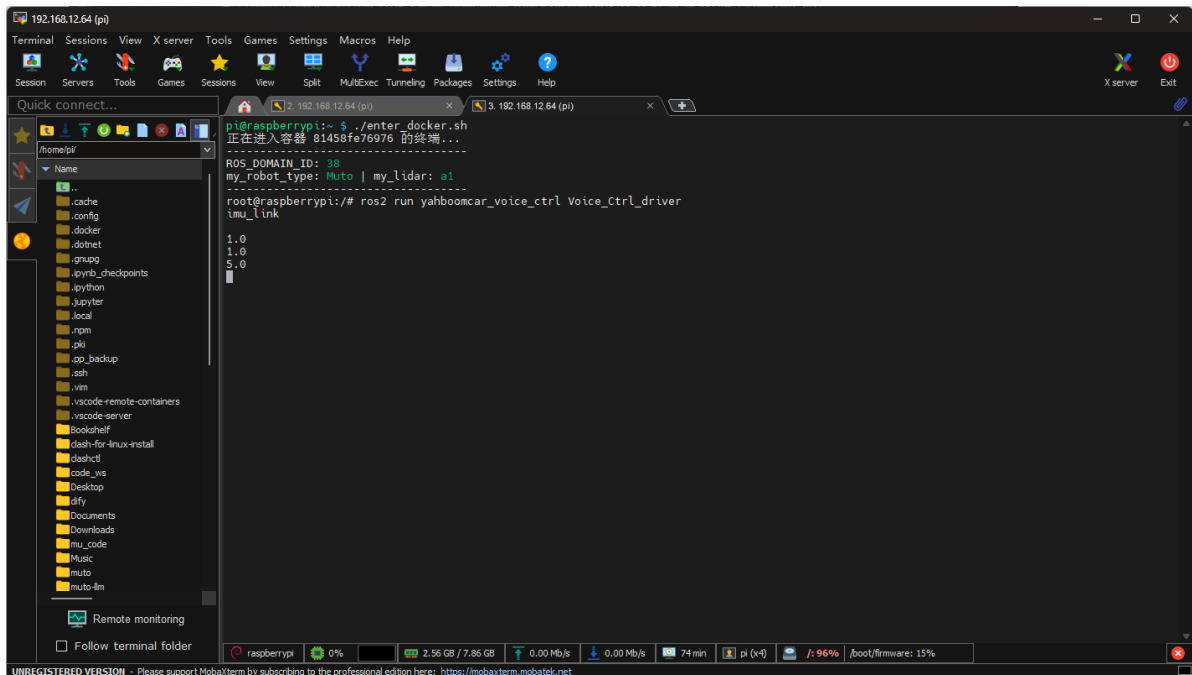
In the first terminal, enter:

```
ros2 launch yahboom_speech speech.launch.py
```



In the second terminal, enter:

```
ros2 run yahboomcar_voice_ctrl Voice_Ctrl_driver
```



After successful startup, you need to say the wake-up word "Hello yahboom".

Muto will reply: "I'm here".

At this point, you can begin voice-controlled movement. The supported command words are as follows (function description on the left, supported command words on the right):

- Motion Control

ROBOT-STOP	ok I am stop
STOP	ok I am stop
ROBOT-SLEEP	ok dormant
GO-AHEAD	ok let's go
BACK	ok i am back
TURN-LEFT	ok i am turning left
TURN-RIGHT	ok i am turning right

4. Core Code

The code in this section is largely similar to `muto_driver.py` in `yahboomcar_bringup`, with the addition of receiving and parsing voice control commands. The following explains how this part is implemented:

```
def voice_callback(self, msg):
    if not isinstance(msg, Int8): return
    speech_r = msg.data
    # Stop
    if speech_r == 2 or speech_r == 0 :
        print("stop!")
        self.execute_vel(0.0, 0.0, 0)
    # Go ahead
    elif speech_r == 4 :
        print("Go ahead!")
        self.speech_and_move(0.5, 0.0, 0)
```

```

# Back off
elif speech_r == 5 :
    print("Back off!")
    self.speech_and_move(-0.5, 0.0, 0)
# Turn left
elif speech_r == 6 :
    print("Turn left")
    self.speech_and_move(0.3, 0.0, 1)
# Turn right
elif speech_r == 7 :
    print("Turn right")
    self.speech_and_move(0.3, 0.0, -1)

```

The command words for this lesson correspond as follows:

- Motion Control

ROBOT-STOP	ok I am stop
STOP	ok I am stop
ROBOT-SLEEP	ok dormant
GO-AHEAD	ok let's go
BACK	ok i am back
TURN-LEFT	ok i am turning left
TURN-RIGHT	ok i am turning right